Emotional Songs

Process Control - Individual project (idea)
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My idea was already submitted and validated by the assistant, but here's still a recap of what it is about!

Idea

A webcam looks at my face and recognises my emotion. When the emotion changes and stays the same for a moment, a matching song starts playing according to the type of emotion detected (for instance a big smile = happiness = *Happy* by Pharrell Williams and so on)

Services

The system has three services. Indeed :

1. **Camera** (sensor): reads the webcam, detects one of *happy, fear, surprise, sad, angry, neutral, etc...* and publishes an event each time the detected emotion changes
2. **Emotion** (processor): ignores detections with low confidence, waits until an emotion has been stable for about 2 seconds, picks a song for it from its own playlist and avoids playing the same song twice in a row. It then publishes which song was selected
3. **"DJ"** (actuator): plays the selected song

Information flow

All services talk through a Mosquitto broker. Topic names follow the structure used in the tutorial (Unit / Intent / Status / Event):

Service	Publishes	Listens to
Camera	`Camera/U/laptop/E/emotion`	–
Emotion	`Emotion/U/laptop/E/songSelected`	Camera event
DJ	`DJ/U/laptop/E/playing`	Emotion event

Tools

Python, `paho-mqtt`, Mosquitto, MQTT Explorer, `deepface` (or `fer`) for emotion detection, `pygame` for playing the songs